Project: Worked examples
Title: Thick-walled cylinder (Lamé)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the radial, hoop and axial stresses at the bore and outer surface of a thick-walled cylinder under internal and external pressure. It also gives von Mises and Tresca stresses, the utilisation and a thin-wall estimate.

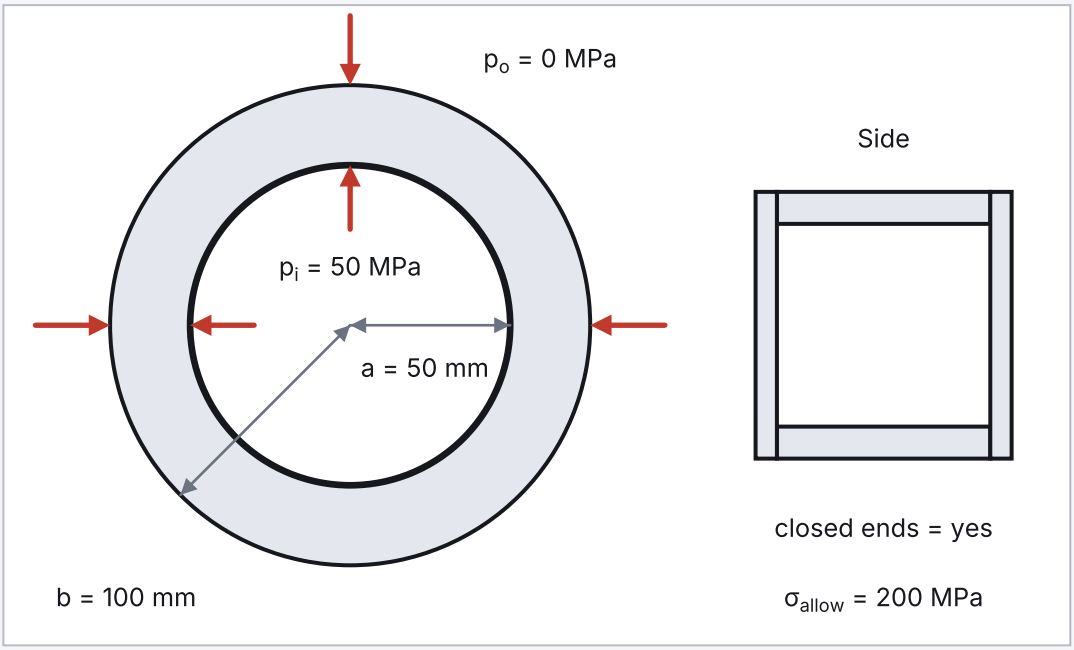


Figure 1 — A thick-walled cylinder in section, bore radius a, outer radius b, pressure inside and out; closed ends carry the pressures axially. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Lamé constants from the bore and outside pressures	$A = (a^2 p_i - b^2 p_o)/(b^2 - a^2)$; $B = a^2 b^2 (p_i - p_o)/(b^2 - a^2)$
Radial and hoop stress at radius r	$\sigma_r = A - B/r^2$; $\sigma_\theta = A + B/r^2$
Axial stress, closed or open ends	$\sigma_z = A$ (closed); $\sigma_z = 0$ (open)
Equivalent stresses from the three principal stresses	von Mises; Tresca = the largest principal difference

Basis: Thick-walled cylinders[1](Ch. 13, Table 13.5, 8th ed.); Stresses in pressurised cylinders[2](§3-14, 10th ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
closed	Closed ends (axial stress carried by the wall)	yes		drawing: DRG-CYL-01 end caps rev A	any
a	Inner radius	50	mm	drawing: DRG-CYL-01 rev A	0.1 … 5000 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
b	Outer radius	100	mm	drawing: DRG-CYL-01 rev A	0.1 … 5000 mm
E	Young's modulus (the elastic collapse pressure under external pressure only)	207	GPa	default	1 … 500 GPa
v	Poisson's ratio (the elastic collapse pressure under external pressure only)	0.3		default	0.1 … 0.49
p_i	Internal pressure	50	MPa	spec: SPEC-PV-2 §3 design pressure	0 … 1000 MPa
p_o	External pressure	0	MPa	assumption: Atmospheric outside.	0 … 1000 MPa
σ_{allow}	Allowable stress	200	MPa	spec: SPEC-PV-2 §5	1 … 3000 MPa

3. Assumptions

A1. The cylinder is long, elastic and isotropic, and its stresses vary with radius only. (Lamé solution, with end effects neglected.)
A2. Closed ends carry the pressure end load as a uniform axial stress. (σ_z = A.)
A3. p_o (External pressure) is taken as 0 MPa. (Atmospheric outside.)

4. Working

[9] $A = (a^2 * p_i - b^2 * p_o) / (b^2 - a^2)$ [1](Table 13.5)
$= ((50\ mm)^2 \times 50\ MPa - (100\ mm)^2 \times 0\ MPa) / ((100\ mm)^2 - (50\ mm)^2)$
$= 16.667\ MPa$
Lamé constant A

[10] $B = a^2 * b^2 * (p_i - p_o) / (b^2 - a^2)$ [1](Table 13.5)
$= (50\ mm)^2 \times (100\ mm)^2 \times (50\ MPa - 0\ MPa) / ((100\ mm)^2 - (50\ mm)^2)$
$= 166670\ MPa \cdot mm^2$
Lamé constant B

[11] $\sigma_{r,a} = A - B / a^2$
$= 16.667\ MPa - 166670\ MPa \cdot mm^2 / (50\ mm)^2$
$= -50\ MPa$
Radial stress at the bore

[12] $\sigma_{t,a} = A + B / a^2$
$= 16.667\ MPa + 166670\ MPa \cdot mm^2 / (50\ mm)^2$
$= 83.333\ MPa$
Hoop stress at the bore

[13] $\sigma_{r,b} = A - B / b^2$
$= 16.667\ MPa - 166670\ MPa \cdot mm^2 / (100\ mm)^2$
$= 0\ MPa$
Radial stress at the outer surface

[14] $\sigma_{t,b} = A + B / b^2$
$= 16.667\ MPa + 166670\ MPa \cdot mm^2 / (100\ mm)^2$
$= 33.333\ MPa$
Hoop stress at the outer surface

[15] $\sigma_z = A$
$= 16.667\ MPa$
$= 16.667\ MPa$
Axial stress (closed ends)

[16] $\sigma_{vm,a}$ = sqrt(((($\sigma_{t,a}$ - $\sigma_{r,a}$)^2 + ($\sigma_{r,a}$ - σ_z)^2 + (σ_z - $\sigma_{t,a}$)^2) / 2)
= sqrt(((83.333 MPa - (-50 MPa))^2 + ((-50 MPa) - 16.667 MPa)^2 + (16.667 MPa - 83.333 MPa)^2) / 2)
= 115.47 MPa

von Mises stress at the bore

[17] $\sigma_{vm,b}$ = sqrt(((($\sigma_{t,b}$ - $\sigma_{r,b}$)^2 + ($\sigma_{r,b}$ - σ_z)^2 + (σ_z - $\sigma_{t,b}$)^2) / 2)
= sqrt(((33.333 MPa - 0 MPa)^2 + (0 MPa - 16.667 MPa)^2 + (16.667 MPa - 33.333 MPa)^2) / 2)
= 28.868 MPa

von Mises stress at the outer surface

[18] $\sigma_{tr,a}$ = max($\sigma_{t,a}$, $\sigma_{r,a}$, σ_z) - min($\sigma_{t,a}$, $\sigma_{r,a}$, σ_z)
= max(83.333 MPa, -50 MPa, 16.667 MPa) - min(83.333 MPa, -50 MPa, 16.667 MPa)
= 133.33 MPa

Tresca stress at the bore

[19] $\sigma_{tr,b}$ = max($\sigma_{t,b}$, $\sigma_{r,b}$, σ_z) - min($\sigma_{t,b}$, $\sigma_{r,b}$, σ_z)
= max(33.333 MPa, 0 MPa, 16.667 MPa) - min(33.333 MPa, 0 MPa, 16.667 MPa)
= 33.333 MPa

Tresca stress at the outer surface

[20] $\sigma_{vm,max}$ = max($\sigma_{vm,a}$, $\sigma_{vm,b}$)
= max(115.47 MPa, 28.868 MPa)
= 115.47 MPa

Maximum von Mises stress

[21] u = $\sigma_{vm,max}$ / σ_{allow}
= 115.47 MPa / 200 MPa
= 0.57735

Utilisation

[22] σ_{thin} = (p_i - p_o) * (a + b) / 2 / (b - a)
= (50 MPa - 0 MPa) × (50 mm + 100 mm) / 2 / (100 mm - 50 mm)
= 75 MPa

Thin-wall hoop estimate at the mean radius

[23] k_{wall} = b / a
= 100 mm / 50 mm
= 2

Wall ratio b/a

5. Results

SYMBOL	VALUE	DESCRIPTION
$\sigma_{t,a}$	83.33 MPa	Hoop stress, bore
$\sigma_{vm,max}$	115.5 MPa	Maximum von Mises (governing)
$\sigma_{r,a}$	-50 MPa	Radial stress, bore
$\sigma_{r,b}$	0 MPa	Radial stress, outer
$\sigma_{t,b}$	33.33 MPa	Hoop stress, outer
σ_z	16.67 MPa	Axial stress
$\sigma_{vm,a}$	115.5 MPa	von Mises, bore
$\sigma_{vm,b}$	28.87 MPa	von Mises, outer

SYMBOL	VALUE	DESCRIPTION
$\sigma_{tr,a}$	133.3 MPa	Tresca, bore
$\sigma_{tr,b}$	33.33 MPa	Tresca, outer
σ_{thin}	75 MPa	Thin-wall hoop estimate
k_{wall}	2	Wall ratio
u	0.58	Utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	von Mises against allowable	115.5 MPa	200 MPa	0.58	

7. Applicability

None.

Defaults used (inputs left empty):
E: not entered, so the default of 207 GPa is used.
v: not entered, so the default of 0.3 is used.

8. Verification

Template mech.thick-cylinder@2.2.1 Content hash e204fc736739a5e7 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.